

FAN YE

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RESEARCH INTERESTS

Low-dimensional topology, especially knot theory, gauge theory and, Floer homology

EMPLOYMENT

Assistant Professor, Peking University	01/2025-now
Simons Collaboration Postdoc, Harvard University, supported by Peter Kronheimer	07/2024-12/2025
Benjamin Peirce Fellow, Harvard University, advised by Peter Kronheimer	07/2022-12/2025

EDUCATION

Ph.D. in DPMMS, University of Cambridge, supervised by Jacob Rasmussen	10/2019-06/2022
Exchange Year, MIT	01/2018-12/2018
B.S. in School of Mathematical Sciences (SMS), Peking University (PKU)	09/2015-06/2019

PHD STUDENTS

Kai Zhu, entered 2025, Peking University
Shengyu Zou, graduated 2026, Peking University, co-supervised with Jiajun Wang

SELECTED TALKS

[2026-07] Short Communication (15 mins), ICM 2026, Philadelphia
[2026-02] Plenary talk, The 21st East Asian Conference on Geometric Topology (EACGT), Seoul

PUBLICATIONS AND PREPRINTS

- [17] **An infinity-categorical TQFT from instantons** *Single author*
submitted, arXiv: 2606.29902.
- [16] **Monopole triangle over integers** *Joint with Haochen Qiu,*
submitted, arXiv: 2606.29882.
- [15] **Instanton 2-torsion and fibered knots** *Joint with Deeparaj Bhat and Zhenkun Li,*
submitted, arXiv: 2512.24206.
- [14] **Singular instanton homology of dual knots** *Single author,*
submitted, arXiv:2511.19883.
- [13] **Instanton dimensions of knot surgeries over arbitrary fields** *Joint with Zhenkun Li,*
submitted, arXiv:2511.17877.
- [12] **Instanton 2-torsion and Dehn surgeries** *Joint with Zhenkun Li,*
submitted, arXiv:2508.03394.
- [11] **2-torsion in instanton Floer homology** *Joint with Zhenkun Li,*
Adv. Math. 472:Paper No. 110289, 55, 2025, arXiv:2405.16252.
- [10] **Knot surgery formulae for instanton Floer homology II: applications**
Joint with Zhenkun Li,
Math. Ann. 391:6291–6371, 2025, arXiv:2209.11018.
- [9] **The guts of nearly fibered knots** *Joint with Zhenkun Li,*
Algebr. Geom. Topol. 26(4): 1385–1393, arXiv:2208.05382.

- [8] **Knot surgery formulae for instanton Floer homology I: the main theorem**
Joint with Zhenkun Li,
Geom. Topol. 29(5): 2269–2342, 2025, *arXiv:2206.10077*.
- [7] **Small Dehn surgery and $SU(2)$** *Joint with John A. Baldwin, Zhenkun Li, and Steven Sivek,*
Geom. Topol. 28(4): 1891–1922, 2024, *arXiv:2110.02874*.
- [6] **$SU(2)$ representations and a large surgery formula** *Joint with Zhenkun Li,*
submitted, arXiv:2107.11005.
- [5] **An enhanced Euler characteristic of sutured instanton homology** *Joint with Zhenkun Li,*
IMRN 2024(4): 2873–2936, 2023, *arXiv:2107.10490*.
- [4] **Instanton Floer homology, sutures, and Euler characteristics** *Joint with Zhenkun Li,*
Quantum Topol. 14 (2): 201–284, 2023, *arXiv:2101.05169*.
- [3] **Sutured instanton homology and Heegaard diagrams**
Joint with John A. Baldwin and Zhenkun Li
Compos. Math. 159(9), 1898–1915, 2023, *arXiv:2011.09424*.
- [2] **Instanton Floer homology, sutures, and Heegaard diagrams** *Joint with Zhenkun Li,*
J. Topol. 15(1): 39–107, 2022, *arXiv:2010.07836*.
- [1] **Constrained knots in lens spaces** *Single author,*
Algebr. Geom. Topol. 23(3): 1097–1166, 2023, *arXiv:2007.04237*.
- [0] **Ph.D. Thesis, New techniques in calculation of sutured instanton Floer homology:**
by Heegaard diagrams, Euler characteristics, and Dehn surgery formulae DOI:10.17863/CAM.85094.

TEACHING EXPERIENCE

Peking University, Undergraduate Geometry	<i>Fall 2026</i>
Harvard Math231A, Algebraic topology	<i>Fall 2024</i>
Harvard Math285Z, Sutured 3-manifolds and Floer homology (topic course)	<i>Spring 2023</i>
Harvard Math230A, Differential geometry	<i>Fall 2023</i>
Harvard Math21A, Multi-variable calculus	<i>Spring 2022</i>
Harvard Math230A, Differential geometry	<i>Fall 2022</i>